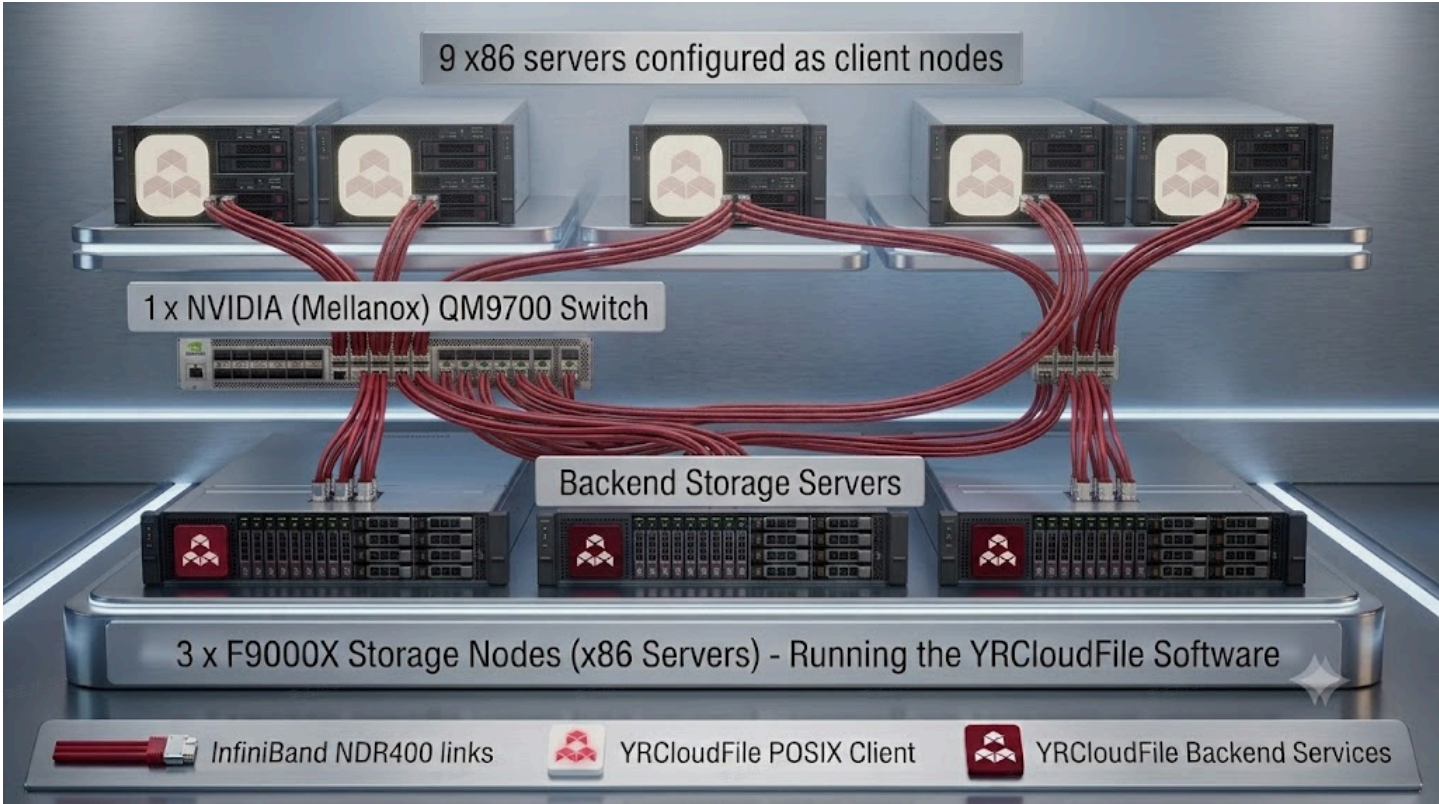


YanRongTech_F9000X_9Clients

1. Hardware

The hardware configuration consists of nine x86 client servers and three backend storage nodes connected to a single central NVIDIA QM9700 switch.



1.1 Client

Client hardware consists of a commercial x86 server featuring a 2P Intel-based motherboard and 256 GB of RAM, configured with default settings. Server-to-client communication is handled via InfiniBand, utilizing two dedicated links allocated exclusively for storage traffic. Client configuration details are summarized in the following table:

Client Node components	Configuration Description	Client Node Count
Client CPU	INTEL(R) XEON(R) PLATINUM 6530 x 2	9
Client Memory	512GB DDR5 (16x 32GB RDIMMs)	
Client OS Boot Drive	480GB SSD x 1	

Client network_type	Mellanox CX7 Single-port NDR400 InfiniBand HCAs x 2	
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1.2 Server

The test environment utilizes a YanRongTech F9000X storage appliance, a commercial x86-based solution running the YRCloudFile distributed file system. The cluster consists of three storage nodes, each equipped with a 480GB SATA SSD for the operating system, sixteen 3.84TB NVMe SSDs for data and metadata, and four single-port ConnectX-7 400Gbps NDR InfiniBand Host Channel Adapters (HCAs) for high-speed connectivity. The servers communicate with the clients over InfiniBand, utilizing four single-port NDR400 adapters to provide a total of four NDR400 links. Detailed server configurations are outlined in the table below:

Server Node components	Configuration Description	Server Node Count
Server CPU	Intel(R) Xeon(R) 6960P x 2	3
Server Memory	3TB DDR5	
Server OS Boot Drive	480GB SATA SSD x 1	
data and metadata drives	Solidigm 3.84TB nvme SSD x 16	
Server network_type	Mellanox CX7 Single-port NDR400 InfiniBand HCAs x 4	

1.3 Switch

The switch is a commercial NVIDIA （Mellanox） QM9700 The client and server is connected to it.

2. Software

2.1 Client

The configuration of the client-side software is shown in the table below:

Software Type	Software Version
Client OS	Ubuntu 24.04 LTS
Client Kernel	6.8.0-60-generic

yrfs-client	YRCloudFile 6.13.4
Mellanox OFED version	MLNX_OFED_LINUX-24.10-4.1.4.0

2.2 Server

The configuration of the Server-side software is shown in the table below:

Software Type	Software Version
YRCloudFile OS	YRCloudFile 6.13.4
Storage Servers OS	Rocky Linux release 9.6 (Blue Onyx)
Storage Servers Kernel	5.14.0-570.17.1.el9_6.x86_64
Mellanox OFED version	MLNX_OFED_LINUX-24.10-4.1.4.0

3. Settings

3.1 Client

The standard yrfs-client (Version: YRCloudFile 6.13.4) is installed on the client using the default installation procedure. yrfs-client is used for the client to access the remote YRCloudFile filesystem.

3.2 Server

For detailed configuration information about storage servers, see the table below:

storage_system	Description
vendor_name	YanRongTech
solution_name	YRCloudFile
model_name	F9000X
version	YRCloudFile 6.13.4
usable_capacity_TB	80

raw_capacity_TB				160				
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3.3 Switch

No tuning applied.